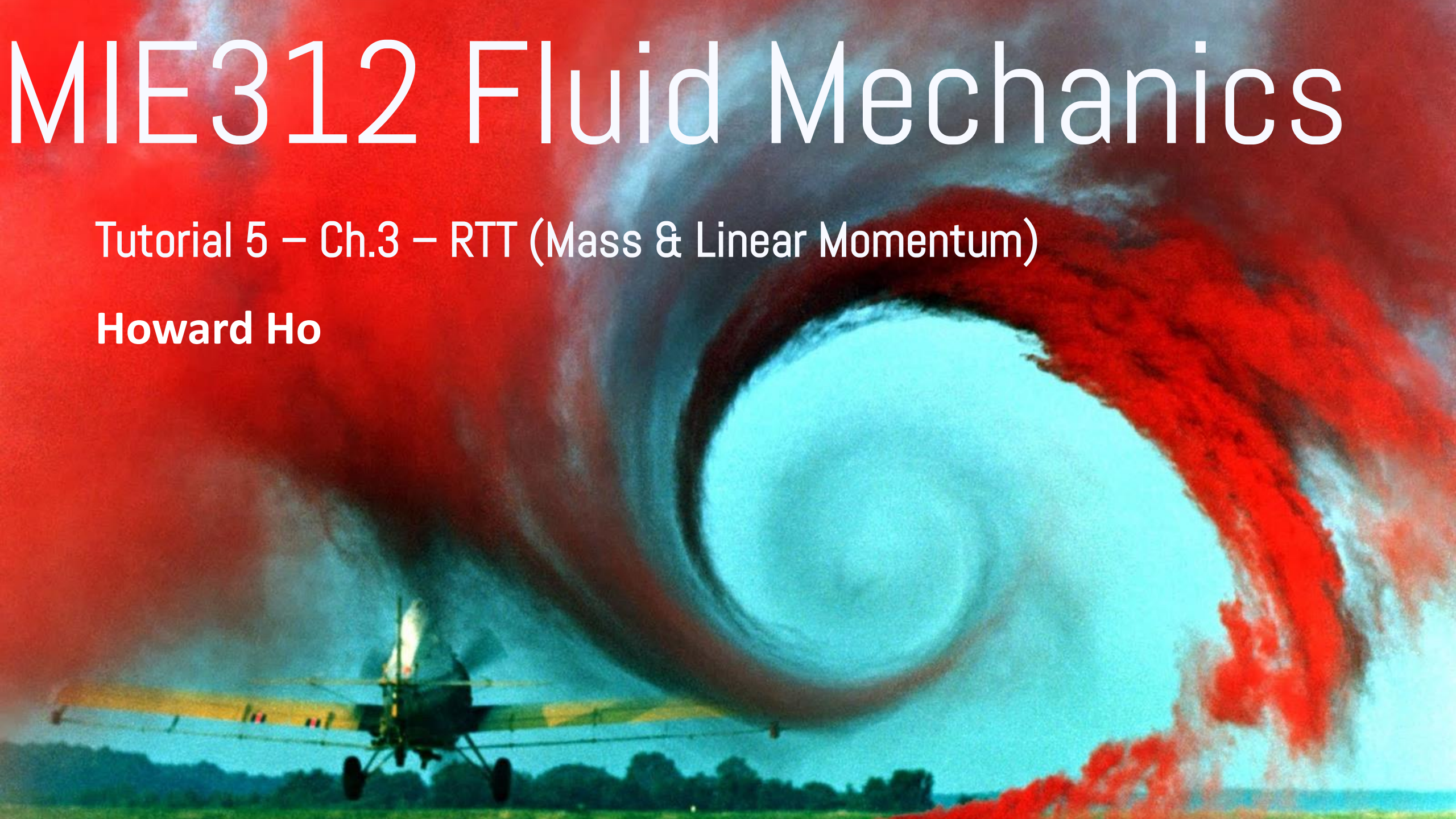
MIE312 Fluid Mechanics

Tutorial 5 – Ch.3 – RTT (Mass & Linear Momentum)

Howard Ho



Tutorial Schedule Changes

Tutorial Week	Topics (2026)
1	Viscosity (Ch.1)
2	Surface Tension (Ch.1)
3	Hydrostatic Pressure (Ch.2)
	Manometer (Ch.2)
4	Mid-Term 1
5	RTT - Mass (Ch.3)
	RTT - Linear Momentum (Ch.3)
6	Bernoulli Equation (Ch.3)
	RTT - Angular Momentum (Ch.3)
X	(Reading Week)
7	Differential Relations (Ch.4)
	Dimensional Analysis (Ch.5)
8	Mid-Term 2
9	Moving CV, non-steady (Ch.3)
	RTT - Energy (Ch.3)
10	Moody Chart - Pipe Flow (Ch.6)
	Head Loss (Ch.6)
11	External Flow (Ch.7)
	Drag Coefficient (Ch.7)

Supplementary Slides

Curved Surface - Pressure (Ch.2)

Bouyancy (Ch.2)

Old

New

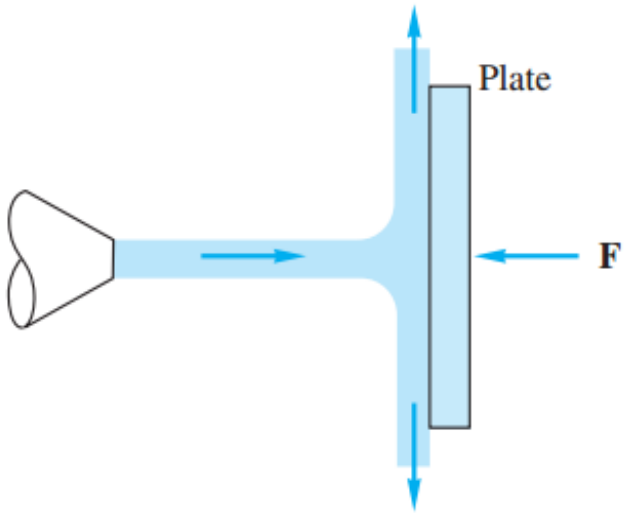
Tutorial Week	Topics (2026)
1	Viscosity (Ch.1)
2	Surface Tension (Ch.1)
3	Hydrostatic Pressure (Ch.2)
	Manometer (Ch.2)
	Curved Surface - Pressure (Ch.2)
4	Mid-Term 1
	Bouyancy (Ch.2)
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	Head Loss (Ch.6)
11	External Flow (Ch.7)
	Drag Coefficient (Ch.7)

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Learning Outcome

- Solve problems with conservation of mass
- Apply Linear Momentum Transport (RTT) equation to calculate forces on
  - Stationary control volumes

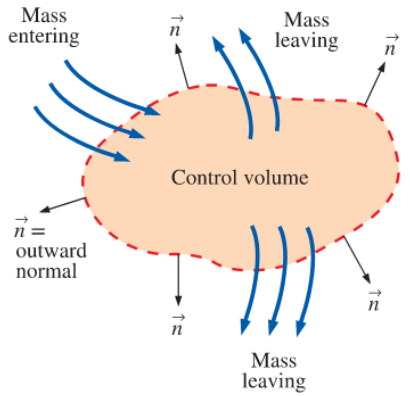






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Reynolds Transport Theorem (RTT)



- Reynolds transport theorem

$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho\,\beta\,d\forall + \int_{CS} \rho\,\beta\,(\vec{V_r}\cdot\hat{n})\,dA$$



$$\beta = \frac{dB}{dm} \qquad \frac{dB_{sys}}{dt} = \frac{dB_{CV}}{dt} - \dot{B}_{in} + \dot{B}_{out}$$

- Conservation of mass

$$B = m, \quad \beta = \frac{dm}{dm} = 1$$

$$0 = \frac{d}{dt}\rho\int d\forall - \dot{m}_{in} + \dot{m}_{out}$$

$$\frac{d}{dt}\rho\int d\forall = \dot{m}_{in} - \dot{m}_{out}$$



- If the flow is steady ($\frac{d}{dt} = 0$),

$$0 = \dot{m}_{in} - \dot{m}_{out}$$

$$\dot{m}_{in} = \dot{m}_{out}$$

Very *fancy* way of saying:
"What goes in must come out"





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Reynolds Transport Theorem (RTT)

- Reynolds transport theorem

$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho\beta \, d\forall + \int_{CS} \rho\beta \left(\overrightarrow{V_r} \cdot \hat{n}\right) dA$$

$$\frac{dB_{sys}}{dt} = \frac{dB_{CV}}{dt} - \dot{B}_{in} + \dot{B}_{out}$$

- Linear Momentum Transport Equation

$$B = m\vec{V} \qquad \beta = \frac{d\, m\vec{V}}{dm} = \vec{V}$$



- Left hand side of equation becomes

$$\frac{d\, m\vec{V}_{sys}}{dt}$$

Recall :

$$\frac{d\vec{V}}{dt} = a$$

$$F = ma$$

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Momentum Transport Equation

- The Linear Momentum Transport Equation has several forms, the original integral form is universal, but pretty scary looking

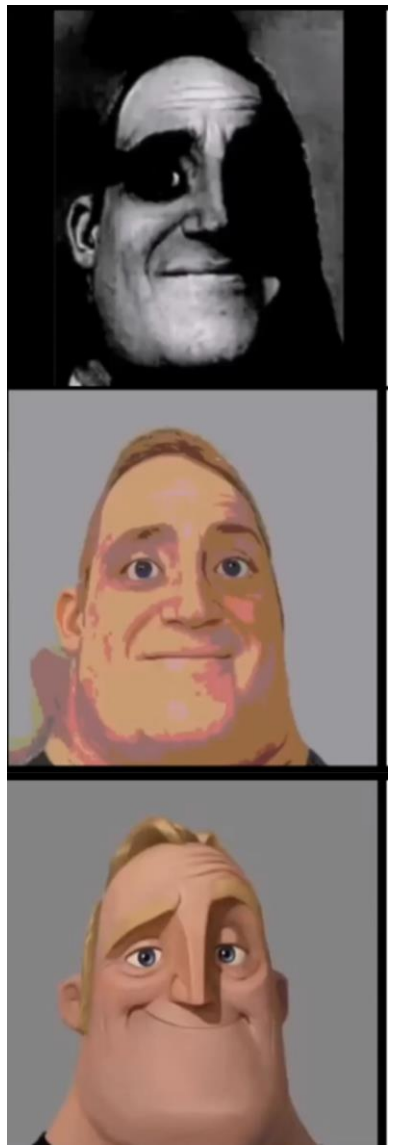
$$\Sigma\vec{F} = \frac{d}{dt}\int_{CV}\rho\vec{V}\,d\forall + \int_{CS}\rho\vec{V}\left(\vec{V_r}\cdot\hat{n}\right)\cdot dA$$

- Most of the time, the control inlet and outlet are 1-D, which makes the equation looks a little nicer

$$\Sigma\vec{F} = \frac{d}{dt}\int_{CV}\rho\vec{V}\,d\forall + \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

- If flow is steady state (e.g. $\frac{d}{dt}\text{anything} = 0$), it gets even better

$$\Sigma\vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$



Conservation of mass:

$$\frac{d}{dt}\rho\int dV = \dot{m}_{in} - \dot{m}_{out}$$

Conservation of linear momentum:

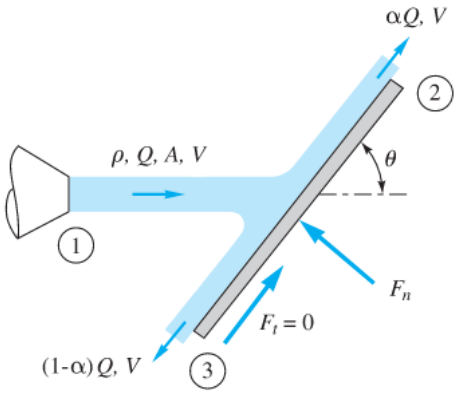
$$m\vec{V}_{in} = m\vec{V}_{out}$$

These two equations are not the same!

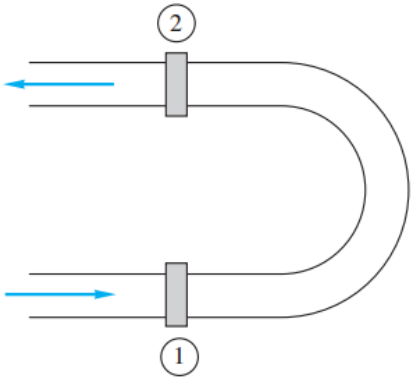
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Problem Types

- Stationary Vane: “classic” scenario, how does the change of flow direction create a force?

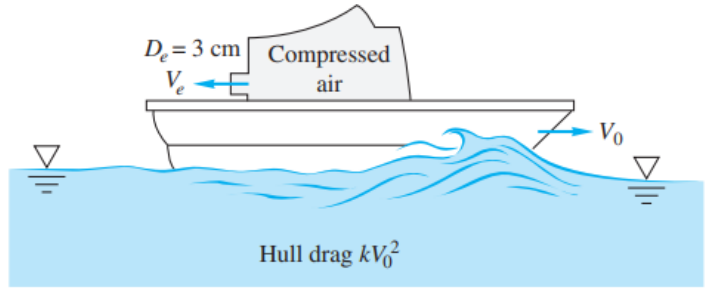


- Pipe Bend: more complex, normally internal flow – Pressure force is present



- Moving Vane: a moving control volume





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Non-moving CV

- When CV is not moving, we can use the simplified form,

$$\Sigma \vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

- But note this is in a vector format (i and j coordinates)

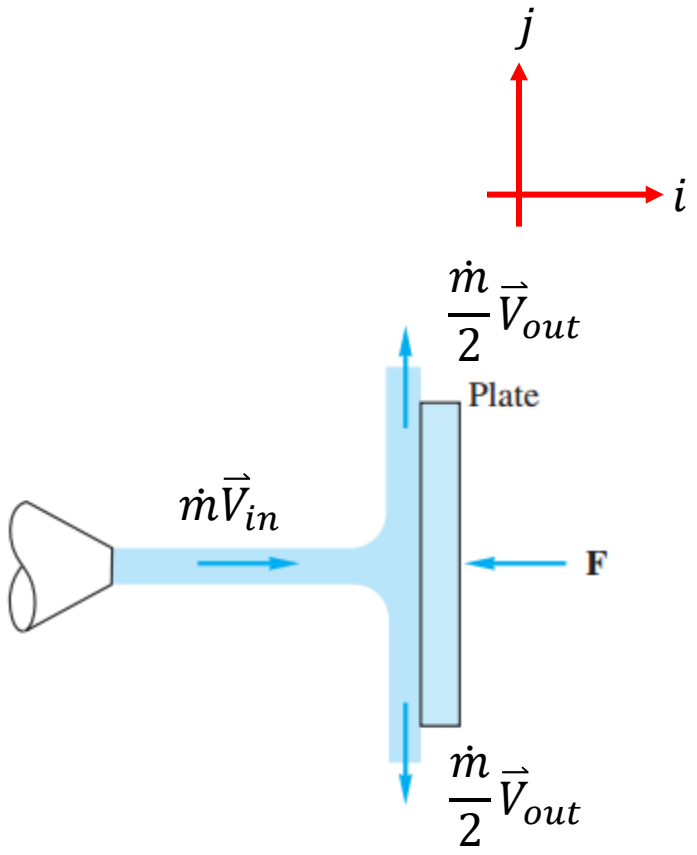
- For example,

$$\Sigma \vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

$$-Fi = \dot{m}\vec{V}_{out}\, j - \dot{m}\vec{V}_{in}\, i$$

$$-Fi = \left(+\frac{\dot{m}}{2}V_{out} - \frac{\dot{m}}{2}V_{out}\right) j - \dot{m}V_{in}\, i$$

$$-Fi = 0\, j - \dot{m}V_{in} i$$



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Steps*

1. Draw your Control Volume (CV): Is CV fixed or moving?

2. Draw the Free-Body Diagram (FBD): Identify all external forces on the CV, e.g. anchoring forces, pressure, gravity and friction

3. Apply Conservation of Mass

4. Apply the Linear Momentum Transport Equation

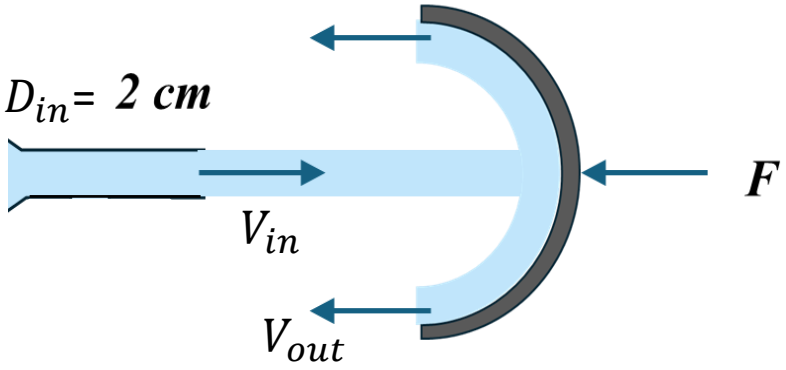
5. Break vector equation in x, y and z components and solve for unknown

5 Steps Guide
1) Draw CV
2) Draw FBD
3) Mass Conservation
4) Momentum Equation
5) Break into x, y and z

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Question 1

Water flows through a circular nozzle at 20 °C, exits into the air as a jet, and strikes a curved plate, which reverses the flow with $A_{out} = \frac{1}{2}A_{in}$. The force required to hold the plate steady is 100 *N*. Assuming steady, frictionless, one-dimensional flow, uniform velocity profile, calculate the velocities at D_{in}.



5 Steps Guide
1) Draw CV
2) Draw FBD
3) Mass Conservation
4) Momentum Equation
5) Break into x, y and z

3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta(\mathbf{V}_r\cdot\mathbf{n})dA$

Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}$; $\dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}$; $\mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out}\dot{m}\,rV_\theta - \sum_{in}\dot{m}\,rV_\theta$

Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Ang. Mom (Stationary):

$\sum\mathbf{M}_{ext} = \frac{\partial}{\partial t}\int_{CV}(\mathbf{r}\times\mathbf{V})\rho d\forall + \int_{CS}(\mathbf{r}\times\mathbf{V})\rho(\mathbf{V}\cdot\mathbf{n})dA$

Momentum Flux (Vanes):

$F_x = \dot{m}(V_{1,x} - V_{2,x})$; $F_y = \dot{m}(V_{1,y} - V_{2,y})$

Energy: $\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$

Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)

Grade Lines: $EGL = HGL + \alpha V^2/2g$

$$\dot{m} = \rho A V$$

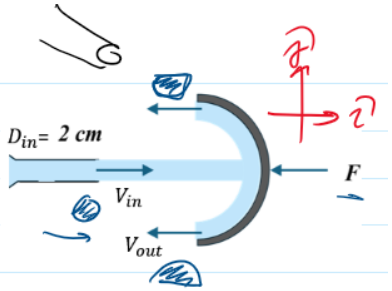
$$\Sigma\vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

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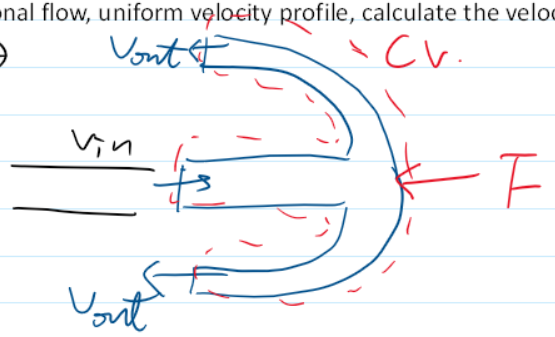
Q1

14 October 2026 15:42

Water flows through a circular nozzle at 20 °C, exits into the air as a jet, and strikes a curved plate, which reverses the flow with $A_{out} = \frac{1}{2}A_{in}$. The force required to hold the plate steady is 100 N. Assuming steady, frictionless, one-dimensional flow, uniform velocity profile, calculate the velocities at D_{in}.



① ③



5 Steps Guide
1) Draw CV
2) Draw FBD
3) Mass Conservation
4) Momentum Equation
5) Break into x, y and z

3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta(\mathbf{V}_r\cdot\mathbf{n})dA$
Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}$; $\dot{m}_{rel} = \rho A V_r$
Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}$; $\mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$
Steady 1-D Ang. Mom.: $M_s = \sum_{out}\dot{m}rV_\theta - \sum_{in}\dot{m}rV_\theta$
Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$
Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int_{CV}\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$
Ang. Mom (Stationary):
$\sum\mathbf{M}_{ext} = \frac{\partial}{\partial t}\int_{CV}(\mathbf{r}\times\mathbf{V})\rho d\forall + \int_{CS}(\mathbf{r}\times\mathbf{V})\rho(\mathbf{V}\cdot\mathbf{n})dA$
Momentum Flux (Vanes):
$F_x = \dot{m}(V_{1,x} - V_{2,x})$; $F_y = \dot{m}(V_{1,y} - V_{2,y})$
Energy: $\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$
Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)
Grade Lines: $EGL = HGL + \alpha V^2/2g$

② Steady-state

$\dot{m}_{in} = \dot{m}_{out}$

$\rho A_{in}\vec{V}_{in} = \rho A_{out}\vec{V}_{out}$

$A_{out} = \frac{1}{2}A_{in}$ (by question)

But 2 A_{out}

$A_{in}\vec{V}_{in} = 2\left(\frac{1}{2}A_{in}\right)\vec{V}_{out}$

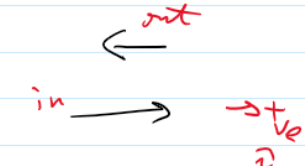
$\therefore \vec{V}_{in} = \vec{V}_{out}$ ③

④ Linear Momentum.

$\sum\vec{F} = 0 + \sum(\dot{m}\vec{V})_{out} - \sum(\dot{m}\vec{V})_{in}$

$-F\hat{\imath} = \dot{m}\vec{V}_{out}\hat{\imath} - \dot{m}\vec{V}_{in}\hat{\imath}$

$(-) = -(2)\left(\frac{\dot{m}}{2}\right)V_{out}\hat{\imath} - \dot{m}V_{in}\hat{\imath}$



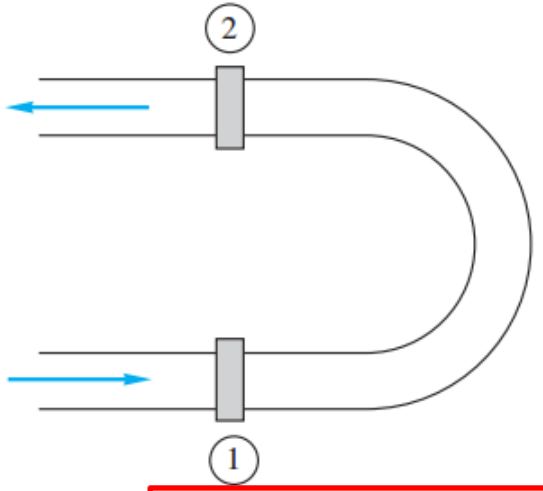
$-F\hat{\imath} = -\dot{m}V_{in}\hat{\imath} - \dot{m}V_{in}\hat{\imath}$

$-100 = -2\rho A_{in}V_{in}^2$

$V_{in} = 12.62\,m/s$ ④

Question 2

Water at 20°C flows through a 5-cm-diameter pipe which has a 180° vertical bend. The total length of pipe between flanges 1 and 2 is 75 cm. When the weight flow rate is 230 N/s, p_1 = 165 kPa, and p_2 = 134 kPa. Neglecting pipe weight, determine the total force which the flanges must withstand for this flow.



5 Steps Guide

1) Draw CV
2) Draw FBD
3) Mass Conservation
4) Momentum Equation
5) Break into x, y and z

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; **HGL:** $HGL = z + P/\gamma$

Isothermal: $P = P_1 \exp[-g\Delta z/RT]$

Troposphere: $P = P_0(1 - \beta z/T_0)^{g/R\beta}$ ($\beta = 0.0065$ K/m)

Hydrostatic: $P = P_0 + \rho gh$

Pascal Law: $F = P \times A$; $P_1 = P_2$

3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV} \rho\beta d\forall + \int_{CS} \rho\beta(\mathbf{V}_r \cdot \mathbf{n})dA$

Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}$; $\dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}$; $\mathbf{U} = \boldsymbol{\omega} \times \mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out} \dot{m} r V_\theta - \sum_{in} \dot{m} r V_\theta$

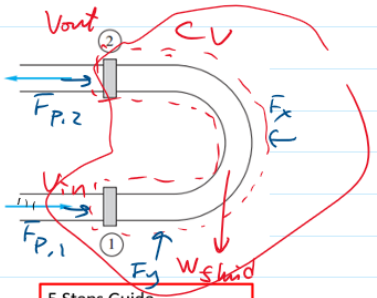
Mass: $\sum \dot{m}_{in} = \sum \dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum \mathbf{F} = \frac{\partial}{\partial t}\int \mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Pressure Forces always act inward into a CV !

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Water at 20°C flows through a 5-cm-diameter pipe which has a 180° vertical bend. The total length of pipe between flanges 1 and 2 is 75 cm. When the weight flow rate is 230 N/s, p_1 = 165 kPa, and p_2 = 134 kPa. Neglecting pipe weight, determine the total force which the flanges must withstand for this flow.



5 Steps Guide
1) Draw CV
2) Draw FBD
3) Mass Conservation
4) Momentum Equation
5) Break into x, y and z

3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta\vec{V}_r\cdot\vec{n}dA$
Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}$; $\dot{m}_{rel} = \rho A V_r$
Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}$; $\mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$
Steady 1-D Ang. Mom.: $M_z = \sum_{out}\dot{m}rV_\theta - \sum_{in}\dot{m}rV_\theta$
Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$
Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$
Ang. Mom (Stationary):
$\sum\mathbf{M}_{xt} = \frac{\partial}{\partial t}\int_{CV}(\mathbf{r}\times\mathbf{V})\rho d\forall + \int_{CS}(\mathbf{r}\times\mathbf{V})\rho(\mathbf{V}\cdot\mathbf{n})dA$
Momentum Flux (Vanes):
$F_x = \dot{m}(V_{1,x} - V_{2,x})$; $F_y = \dot{m}(V_{1,y} - V_{2,y})$
Energy: $\frac{p_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{p_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$
Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)
Grade Lines: $EGL = HGL + \alpha V^2/2g$

① ✓ ② Pressure force always act inward into the CV. (ignores direction of flow).

③ D = 5cm or 0.05m

$\dot{m}_{in} = \dot{m}_{out}$

$\rho A_{in}\vec{V}_{in} = \rho A_{out}\vec{V}_{out}$ (not very useful ...)

$\dot{W}\left(\frac{dW}{dt}\right)$ volume

To get $\dot{W}$, we need $\dot{\forall}$

[N/s] [m^3/s]

$\dot{\forall}$ ρ [m^3/s · kg/m^3]

$\dot{m}$ [kg/s]

g

$\dot{m}g = \dot{w}$ 230 / 9.81

$\dot{m}$ = 23.45 kg/s

ρ_{water} = 1000 kg/m^3

$\dot{m} = \rho A\vec{V}$

$23.45 = 1000(\pi r^2)\vec{V}$ r = 0.05/2

V = 12 m/s ($V_{in} = V_{out}$)

D → R

The reason we subtract P_{atm} from P_1 and P_2 is that we use P_{gage} for calculating hydrostatic pressure forces. The question gave us $P_{absolute}$ (they didn't state it explicitly in the question, but the solution shows it). As for the reason why we use P_{gage} is that P_{atm} acts on every surface of the CV (e.g. it's on both inside and outside surface of the pipe, therefore they cancel out), and we have to reflect that with our calculations (in exams the pressure type will be stated explicitly)

④ Linear Momentum.

$\vec{z}$: $\sum F_x = \sum\dot{m}\vec{V}_{out} - \sum\dot{m}\vec{V}_{in}$

$P_{gage,1}$ P_{atm} P_{atm}

$-F_x + (165-100)(1000)A_1 + (134-100)(1000)A_2 = RHS$

100 kPa (Patm)

$-F_x + 127.6 + 66.8 = -\dot{m}V_{out} - \dot{m}V_{in}$ ✓ ✓

$-F_x + 194.4 = -2(23.45)(12)$

$F_x = 757$ N ←

→ $\vec{z}$

$\uparrow$ $\sum F_y = \sum(\dot{m}\vec{V}_{out}) - \sum(\dot{m}\vec{V}_{in})$ (both = 0)

$F_y - W_f = 0$

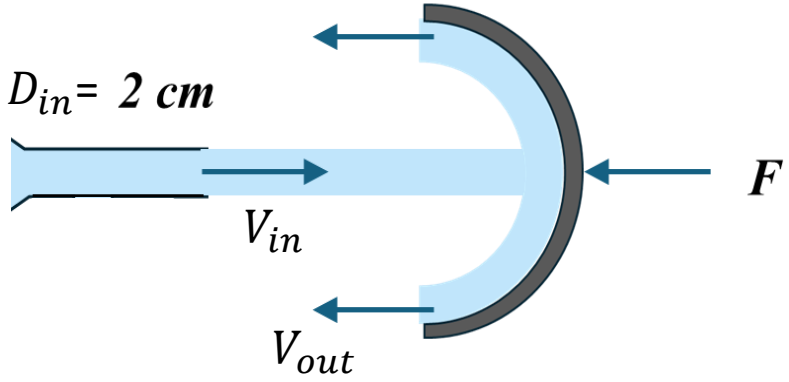
$F_y = (\pi r^2)(L)\rho g$ (L given)

$F_y = 14$ N ↑

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Question 1

Water flows through a circular nozzle at 20 °C, exits into the air as a jet, and strikes a curved plate, which reverses the flow with $A_{out} = \frac{1}{2}A_{in}$. The force required to hold the plate steady is 100 N. Assuming steady, frictionless, one-dimensional flow, uniform velocity profile, calculate the velocities at D_{in}.



Mass Conservation:

$$\dot{m}_{out} = \dot{m}_{in}$$

$$\rho A_{in} V_{in} = \rho A_{out} V_{out}$$

$$\rho A_{in} V_{in} = 2\,\rho(\frac{1}{2}A_{in})V_{out}$$

Which gives,

$$V_{in} = V_{out}$$

Linear Momentum Transport Equation:

$$\vec{F} = \dot{m}\,\vec{V}_{out} - \dot{m}\,\vec{V}_{in}$$

$$-Fi = \dot{m}\,\vec{V}_{out}\,i - \dot{m}\,\vec{V}_{in}\,i$$

$$-Fi = \left(-2\frac{\dot{m}}{2}V_{out}\right)i - \dot{m}\,V_{in}\;\; i$$

$$-F\,i = -\dot{m}\,V_{in}\,i\; - \dot{m}\,V_{in}\,i$$

$$-100 = -2\rho A_{in} V_{in}^2$$

$$V_{in} = 12.62\ m/s$$

$$\dot{m} = \rho A V$$

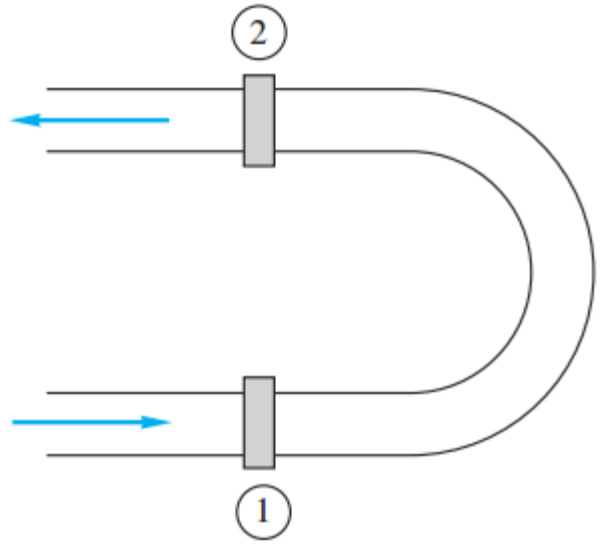
$$\Sigma\vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

4 Steps Guide

1) Draw CV
2) Draw FBD
3) Momentum Equation
4) Break into x, y and z

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Question 2



P3.43 Water at 20°C flows through a 5-cm-diameter pipe which has a 180° vertical bend, as in Fig. P3.43. The total length of pipe between flanges 1 and 2 is 75 cm. When the weight flow rate is 230 N/s, p_1 = 165 kPa, and p_2 = 134 kPa. Neglecting pipe weight, determine the total force which the flanges must withstand for this flow.

Solution: Let the CV cut through the flanges and surround the pipe bend. The mass flow rate is (230 N/s)/(9.81 m/s^2) = 23.45 kg/s. The volume flow rate is Q = 230/9790 = 0.0235 m^3/s. Then the pipe inlet and exit velocities are the same magnitude:

$$V_1 = V_2 = V = Q/A = \frac{0.0235\ \text{m}^3/\text{s}}{(\pi/4)(0.05\ \text{m})^2} \approx 12.0\ \frac{\text{m}}{\text{s}}$$

Subtract pa everywhere, so only p1 and p2 are non-zero. The horizontal force balance is:

$$\sum F_x = F_{x,\text{flange}} + (p_1 - p_a)A_1 + (p_2 - p_a)A_2 = \dot{m}_2 u_2 - \dot{m}_1 u_1$$

$$= F_{x,\text{fl}} + (64000)\frac{\pi}{4}(0.05)^2 + (33000)\frac{\pi}{4}(0.05)^2 = (23.45)(-12.0 - 12.0\ \text{m/s})$$

or: $F_{x,\text{flange}} = -126 - 65 - 561 \approx \mathbf{-750\ N}$ *Ans.*

The total x-directed force on the flanges acts to the left. The vertical force balance is

$$\sum F_y = F_{y,\text{flange}} = W_{\text{pipe}} + W_{\text{fluid}} = 0 + (9790)\frac{\pi}{4}(0.05)^2(0.75) \approx \mathbf{14\ N} \quad \textit{Ans.}$$

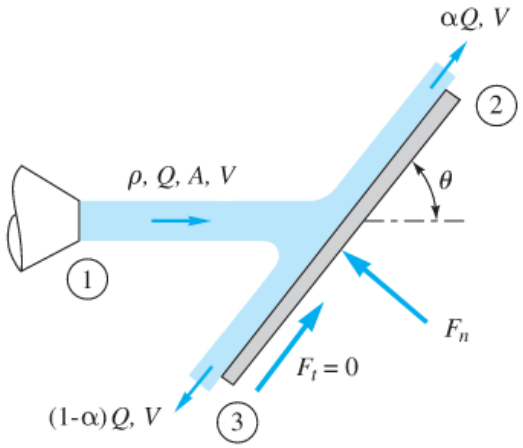
Clearly the fluid weight is pretty small. The largest force is due to the 180° turn.

4 Steps Guide
1) Draw CV
2) Draw FBD
3) Momentum Equation
4) Break into x,y and z

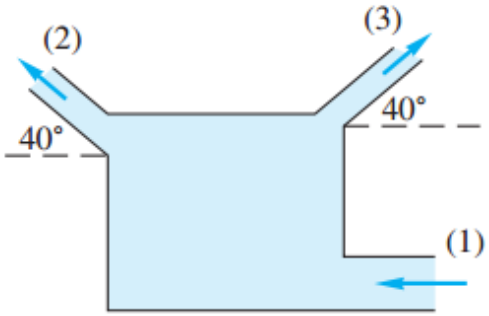
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Practice Problems

P3.46 When a jet strikes an inclined plate, it breaks into two jets of equal velocity V but unequal fluxes αQ at (2) and (1 – α)Q at (3), as shown. Find α, assuming that the tangential force on the plate is zero. Why doesn't the result depend upon the properties of the jet flow?

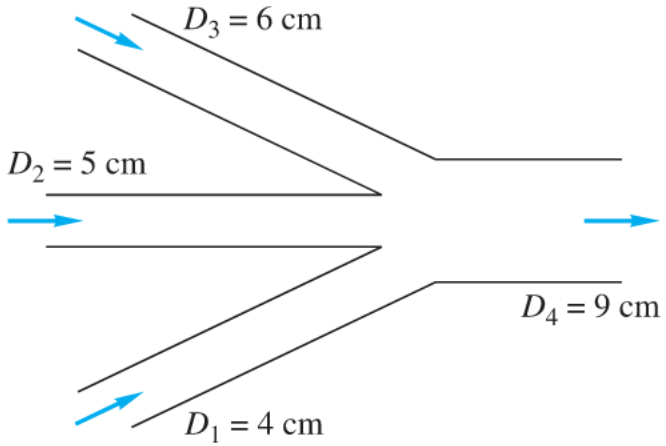


P3.63 Water flows steadily through the box in Fig. P3.63. Average velocity at all ports is 7 m/s. The vertical momentum force on the box is 36 N. What is the inlet mass flow?



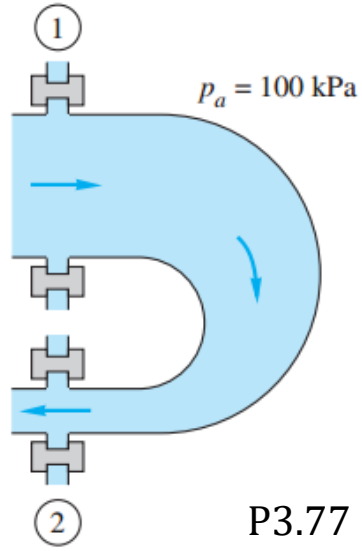
P3.63

Three pipes steadily deliver water at 20°C to a large exit pipe in Fig. P3.8. The velocity $V_2 = 5$ m/s, and the exit flow rate $Q_4 = 120$ m^3/h. Find (*a*) V_1, (*b*) V_3, and (*c*) V_4 if it is known that increasing Q_3 by 20 percent would increase Q_4 by 10 percent.



P3.8

P3.77 Water at 20°C flows steadily through a reducing pipe bend, as in Fig. P3.77. Known conditions are $p_1 = 350$ kPa, $D_1 = 25$ cm, $V_1 = 2.2$ m/s, $p_2 = 120$ kPa, and $D_2 = 8$ cm. Neglecting bend and water weight, estimate the total force that must be resisted by the flange bolts.

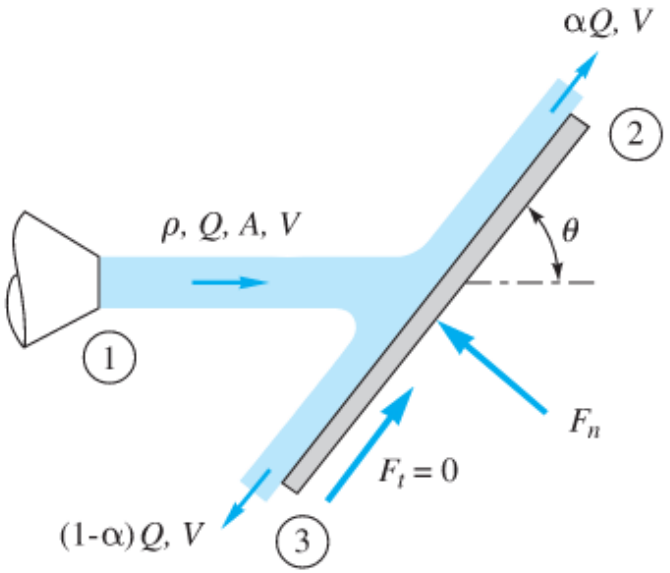


P3.77

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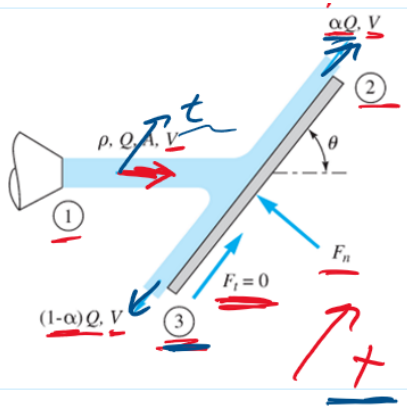
Practice Problem Solution

P3.46 When a jet strikes an inclined plate, it breaks into two jets of equal velocity V but unequal fluxes αQ at (2) and $(1-\alpha)Q$ at (3), as shown. Find α, assuming that the tangential force on the plate is zero. Why doesn't the result depend upon the properties of the jet flow?



$$\Sigma F = \dot{m}V_{out} - \dot{m}V_{in}$$

P3.46 When a jet strikes an inclined plate, it breaks into two jets of equal velocity V but unequal fluxes αQ at (2) and $(1-\alpha)Q$ at (3), as shown. Find α, assuming that the tangential force on the plate is zero. Why doesn't the result depend upon the properties of the jet flow?



* Tangential dir^n

$\Sigma F_t = 0 = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$ (+)

$0 = \dot{m}_2 V - \dot{m}_3 V - \dot{m}_1 V\cos\theta$

$\dot{m} = \rho Q$

$0 = \rho\alpha Q V - \rho(1-\alpha)QV - \rho QV\cos\theta$

$0 = \alpha - (1-\alpha) - \cos\theta$

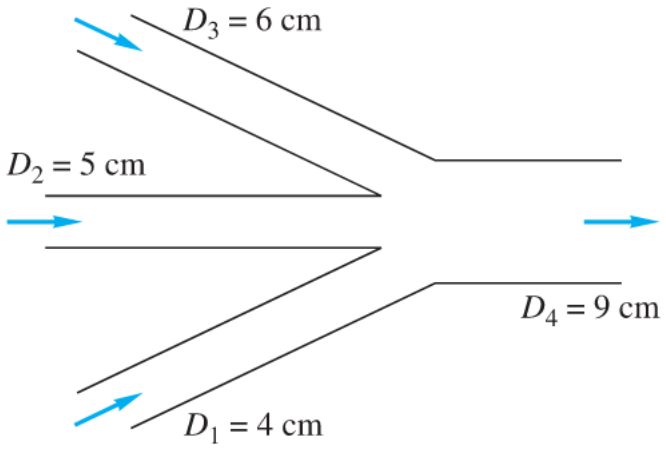
$\alpha = \frac{1+\cos\theta}{2}$ //

mass flow rate cancels out.

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Practice Problem Solutions

Three pipes steadily deliver water at 20°C to a large exit pipe in Fig. P3.8. The velocity $V_2 = 5$ m/s, and the exit flow rate $Q_4 = 120$ m^3/h. Find (*a*) V_1, (*b*) V_3, and (*c*) V_4 if it is known that increasing Q_3 by 20 percent would increase Q_4 by 10 percent.



P3.8

Solution: (a) For steady flow we have $Q_1 + Q_2 + Q_3 = Q_4$, or

$$V_1A_1 + V_2A_2 + V_3A_3 = V_4A_4 \qquad (1)$$

Since $0.2Q_3 = 0.1Q_4$, and $Q_4 = (120\ \text{m}^3/\text{h})(1\ \text{h}/3600\ \text{s}) = 0.0333\ \text{m}^3/\text{s}$,

$$V_3 = \frac{Q_4}{2A_3} = \frac{(0.0333\ \text{m}^3/\text{s})}{\frac{\pi}{2}(0.06^2)} = \mathbf{5.89\ m/s} \quad \textit{Ans.}\ (b)$$

Substituting into (1),

$$V_1\left(\frac{\pi}{4}\right)(0.04^2) + (5)\left(\frac{\pi}{4}\right)(0.05^2) + (5.89)\left(\frac{\pi}{4}\right)(0.06^2) = 0.0333 \qquad \mathbf{V_1 = 5.45\ m/s} \quad \textit{Ans.}\ (a)$$

From mass conservation, $Q_4 = V_4A_4$

$$(0.0333\ \text{m}^3/\text{s}) = V_4(\pi)(0.06^2)/4 \qquad \mathbf{V_4 = 5.24\ m/s} \quad \textit{Ans.}\ (c)$$

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Practice Problem Solutions

P3.63 Water flows steadily through the box in Fig. P3.63. Average velocity at all ports is 7 m/s. The vertical momentum force on the box is 36 N. What is the inlet mass flow?

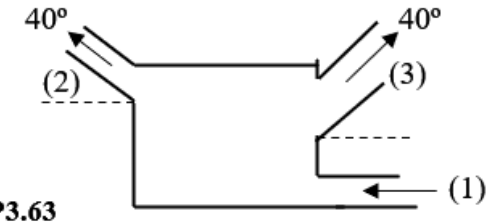


Fig. P3.63

Solution: We don't need water density, just mass flow. Continuity requires that $\dot{m}_1 = \dot{m}_2 + \dot{m}_3$, and then, with a control volume around the entire system, steady vertical momentum requires that

$$\sum F_y = \dot{m}_2 v_2 + \dot{m}_3 v_3 - \dot{m}_1 v_1 = \dot{m}_2 V \sin 40° + \dot{m}_3 V \sin 40° - \dot{m}_1 (0) =$$
$$(\dot{m}_2 + \dot{m}_3) V \sin 40° = \dot{m}_1 (7 m/s) \sin 40° = \dot{m}_1 (4.50 m/s) = 36 N$$
$$\text{Solve for} \quad \dot{m} = \mathbf{8.0\ kg/s} \quad \textit{Ans.}$$

P3.77 Water at 20°C flows steadily through a reducing pipe bend, as in Fig. P3.77. Known conditions are $p1 = 350$ kPa, $D1 = 25$ cm, $V1 = 2.2$ m/s, $p2 = 120$ kPa, and $D2 = 8$ cm. Neglecting bend and water weight, estimate the total force which must be resisted by the flange bolts.

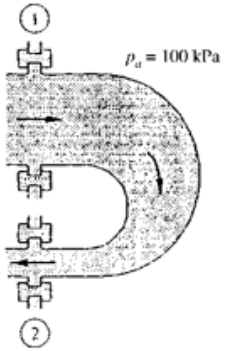


Fig. P3.76

Solution: First establish the mass flow and exit velocity:

$$\dot{m} = \rho_1 A_1 V_1 = 998 \left(\frac{\pi}{4}\right)(0.25)^2 (2.2) = 108 \ \frac{kg}{s} = 998 \left(\frac{\pi}{4}\right)(0.08)^2 V_2, \quad \text{or} \quad V_2 = 21.5 \ \frac{m}{s}$$

The CV surrounds the bend and cuts through the flanges. The force balance is

$$\sum F_x = -F_{bolts} + p_{1,gage} A_1 + p_{2,gage} A_2 = \dot{m}_2 u_2 - \dot{m}_1 u_1, \quad \text{where } u_2 = -V_2 \text{ and } u_1 = V_1$$

$$\text{or} \quad F_{bolts} = (350000 - 100000)\frac{\pi}{4}(0.25)^2 + (120000 - 100000)\frac{\pi}{4}(0.08)^2 + 108(21.5 + 2.2)$$
$$= 12271 + 101 + 2553 \approx \mathbf{14900\ N} \quad \textit{Ans.}$$

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